Jeanne M. Holm Center

Guide to
Air Force Employment Exercise (AFEX)
Theater Airpower Visualization (TAV) Wargame



Air Force ROTC
Officer Training School
Civilian Acculturation & Leadership Training

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Air Force Employment Exercise (AFEX)

Introduction to AFEX



Welcome to AFEX. AFEX is one of several student wargames developed with Theater Airpower Visualization (TAV) software. TAV presents air and space power scenarios that are hypothetical or based on historical events. These sequences give students the means by which to plan an air campaign but also adjust their strategy in real time as the scenario plays out. Several other TAV scenarios may be made available to students including illustrations of Operation Desert Storm, OAF Kosovo, and DHS operations in the aftermath of Hurricane Katrina. *(To obtain copies of additional scenarios, contact the Holm Center Curriculum department.)*

For the purpose of your warfare studies, you will use AFEX, a scenario which depicts a fictitious conflict between the United States and Cuba (which is receiving financial and military backing from other nations). As a capstone event, your flight will plan and execute an air campaign whereby the United States will attempt to conduct a Noncombatant Evacuation Operation (NEO) out of Guantanamo Bay Cuba.

AFEX Planning

To assist you with campaign planning, the AFEX guide includes an Execution Order (EXORD). Found on pages 22-23 (Appendix 1), the EXORD outlines the regional situation that is driving the crisis with Cuba and provides you the Combined Forces Air Component Commander's (CFACC) tasks. In addition, Appendix 2 on pages 24-29 provides a suggested team breakout along with a list of assets and team tasks. However, the EXORD and team assignments are only a launch pad for more detailed planning. Teams will need to develop more detailed objectives that are SMART - specific, measurable, attainable, realistic, and timely.

While the AFEX guide provides you with the "WHAT" (what's the scenario, what are the tasks, etc.), your challenge will be to determine the "HOW" (how to accomplish the objectives, how to effectively package aircraft and ordnance for greatest effect, etc.) The mechanism for accomplishing air campaign tasks and objectives is the AFEX TAV engine.

The best way to become familiar with the AFEX TAV engine is simply to experiment. In fact, we recommend teams practice an entire AFEX scenario, time permitting, prior to playing the game as a capstone event. This will not only give you a better understanding of game mechanics but will also provide the team intelligence preparation of the battlespace (IPB). This IPB will help teams better understand the enemy's capabilities and allow them to more effectively plan.

AFEX Evaluation

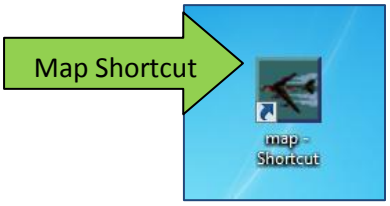
AFEX is a both a test of one's understanding of air and space employment as well as an exercise in critical thinking, communication, and teamwork. To be successful at AFEX, one must have a basic grasp of weapon system capabilities, principles of warfare, force packaging, and air and space power functions. In addition, AFEX will challenge teams to develop a strategy and work together under stressors and time constraints to carry out their plan.

Via the Victory Condition Report (accessible under the View Menu on the host computer), the AFEX engine provides objective feedback on several measures of merit to include target effectiveness, aircraft losses, rescued crews, air refueling, ISR hits, etc. However, the instructor and team must also determine subjective assessment of success based on the team's SMART objectives going into game play and how well members lead, follow, and communicate. The Victory Condition Report should not be considered a means of determining success or failure but merely a way of providing data points.

AFEX General Set-Up Instructions

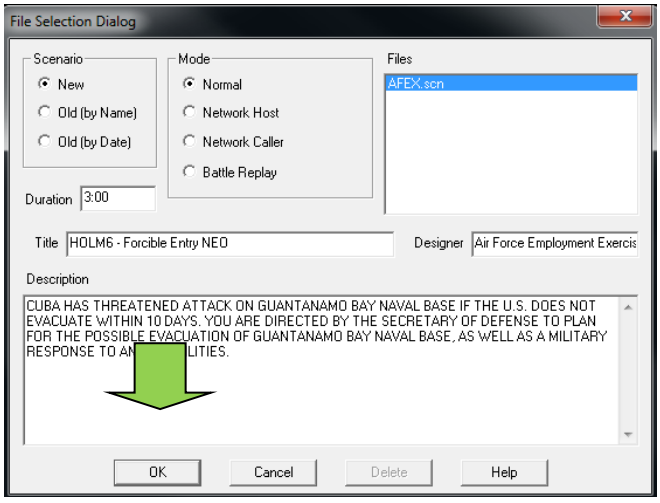
Your scenario can be practiced alone versus the computer (no local area network or internet required) OR as a team using Network Play versus the computer. To play AFEX as a team, you'll need up to seven (7) laptop or desktop computers and a router, MiFi/WiFi access, or local area network. We recommend instructors or students save the AFEX DVD file or Zip file to computer hard drives. Periodic file updates may be posted to the Holmcenter.com website. See Appendix 4 on page 33 for additional set-up considerations.

For purposes of orienting yourself to TAV, begin practicing as an individual.

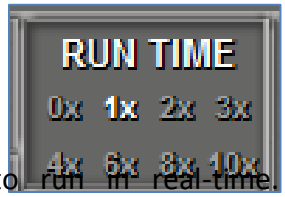


Running TAV (Individual)

1. Launch TAV by clicking on the "map" or "AFEX" application icon on your disk or desktop, identifiable as a B-52.



2. When prompted by the popup Fire Selection Dialog, stay in "Normal" mode. Ensure the "AFEX" option is highlighted. Then click "OK"

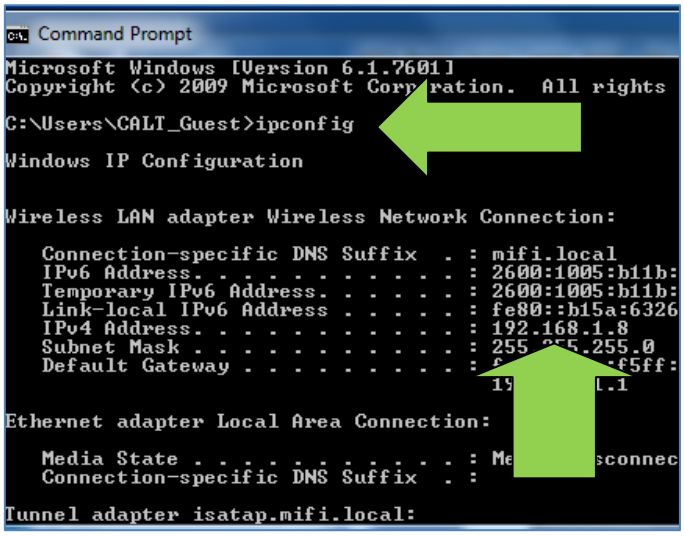


3. To select a specific run time speed, click on the desired factor in the Flight Information area. Selecting **1x** will cause the program to run in real-time. Selecting **0x** or pressing the Space Bar Hot Key will pause the program. (While in **0x** run time, you can reposition or preposition assets.) While in normal mode, you can also select speeds higher than real-time up to **10x.**

4. Each scenario has a duration measured in hours, minutes, and seconds. The current date and time is displayed in the middle of the Flight Information Area (see page 8) and the remaining time is displayed below that. When the remaining time reaches zero, then the scenario is automatically stopped and the victory status displayed.

Running TAV (Team Network Play)

There are at least three ways to set up network play.

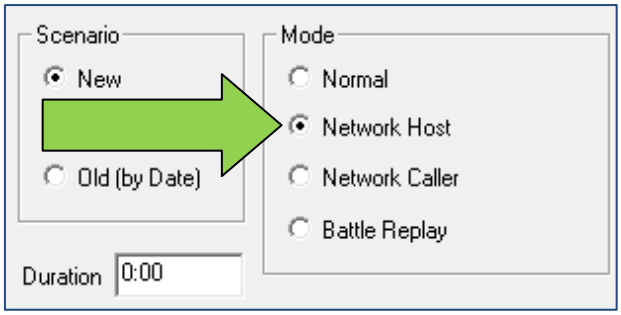


OPTION 1: Set up your own stand-alone network with a router box and connect each terminal to the router. This approach may be best if you anticipate "firewall" issues.

OPTION 2: Use WiFi or MiFi to connect your computers. This approach may also work well if you anticipate "firewall" issues.

OPTION 3: Connect your computers over a local area network.

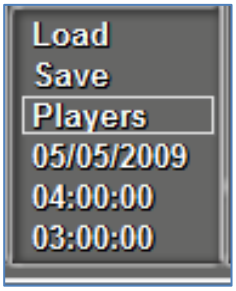
Team Network Play requires a "Network Host" computer and six "Network Caller" computers.



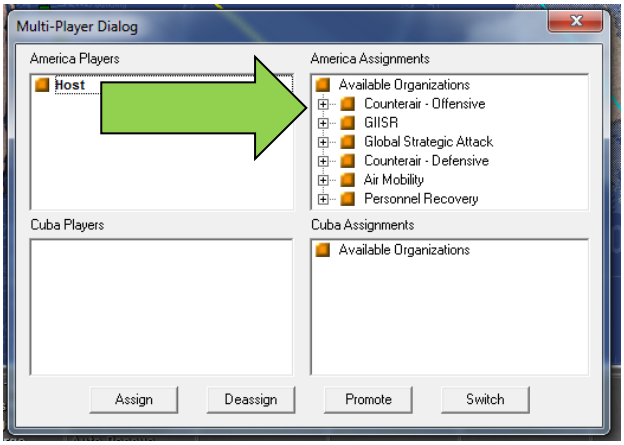
1. The first computer to log in will be the host computer. Before initiating the game, the host will need to determine the IP address for his or her computer. To obtain the IP address, click start, select programs, accessories, and then command prompt. Type in "**ipconfig**" with no spaces and hit enter. Another approach is to search for your command prompt by typing in "Command Prompt." Have team members write down the IP address. An example of an IP address format is 192.168.1.8.

2. The host computer should then launch TAV by clicking on the "map" or "AFEX" application icon on his or her disk or desktop. When prompted by the popup Fire Selection Dialog, select "Network Host" mode and AFEX from the files dialogue box. Then click "OK."

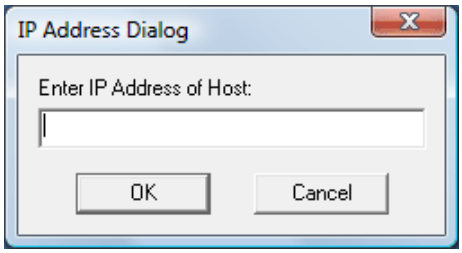
The host player will then prepare to assign teams by clicking on "Players" in the Run Time box.



Initially, the roles under America assignments will be blank. However, as America players log in, the host will be able to assign available organizations to each player by highlighting the America player

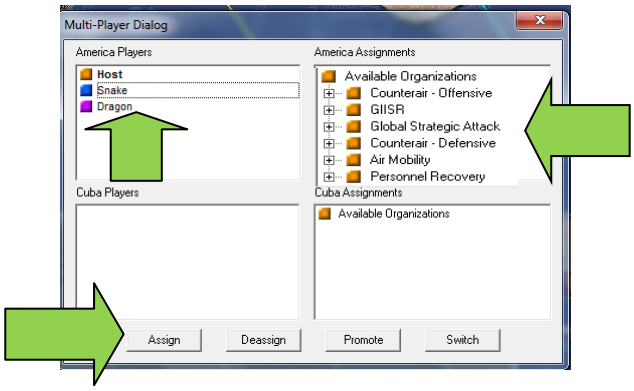


As team members log in, they should select "Network Caller." Members will type in their designated team role and check the box to play on the same side as the host player. Then they will be prompted to enter the IP address of the host.



Once players are assigned to roles, the host will need to assign available organizations (aircraft assets) to each team in keeping with the flight's overall strategy.

In the example to the right, a team has logged in using call sign "Snake" and the host is assigning Global Strategic Attack assets by highlighting the team and the available organization and clicking on "**Assign**." Note that the squadrons automatically assigned under the main organizational headings may be reapportioned to other teams IAW the overall air campaign plan.

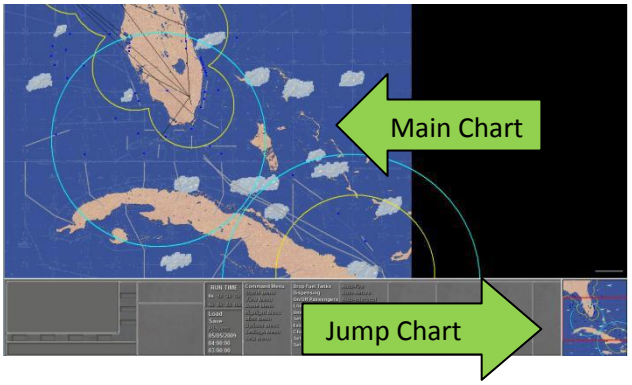


When the host has assigned all assets, the team should be ready to begin play. Note that the Network Play mode is restricted to three run time speed options: **0x**, **1x**, and **2x**.

Navigating TAV

Main Chart

The Main Chart has 4 zoom levels. These can be toggled by using the **1**, **2**, **3**, and **4** hot keys. If your mouse has a scroll wheel, then this can also be used to zoom in and out on the map.



The Main Chart can also be scrolled by moving the mouse to the edge of the screen. This will cause the Main Chart to scroll in the direction of the mouse.

In addition, if you hold down the **Shift** key and left-click on the map as you hold down the mouse button, as you move the cursor around the map you will automatically scroll the map.

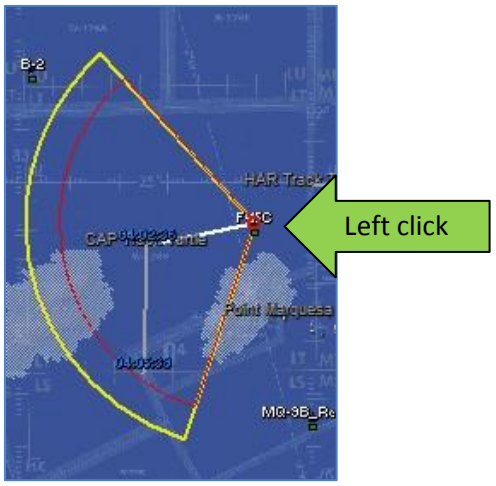
Jump Chart

The Jump Chart is a smaller area in the lower left-hand corner of the screen that shows the entire battle area. Flights of the host side are blue dots and flights of the opposing side are red dots. Clicking with the left mouse button in the Jump Chart will cause the Main Chart to jump to that location.

Selecting a Flight

If you click on a flight with the left-mouse button, then it becomes **selected** and is shown in the Flight Information area. On the Main Chart, the selected flight is shown surrounded by a red circle.

Common Range Indicators



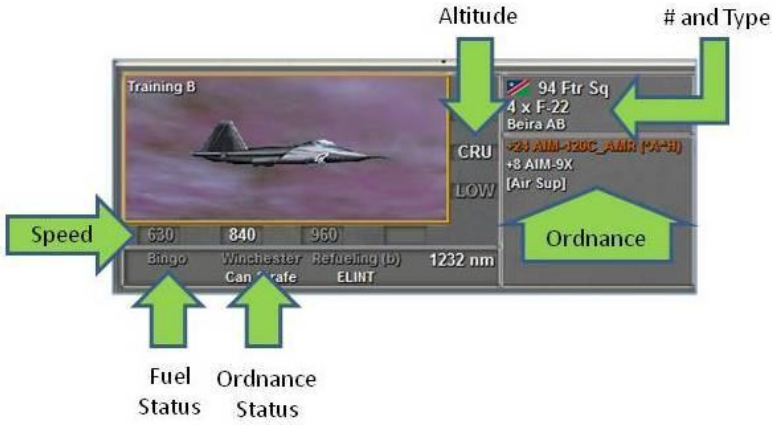
When you click on a flight, the aircraft's range indicators will be visible. Below are the more common types of range indications provided for flights. See page 21 for a complete list of range indicators.

Blue circles indicate current visibility or what the flight can see visually.

Yellow wedges show the maximum range of any on-board radar that the aircraft in the flight carries. Enemy flights can potentially be detected within the area of this wedge.

Bright red wedges show the maximum range of the currently selected ordnance. Flights can potentially fire upon enemy flights within this wedge.

Flight Information Area



Once you click on a flight, the flight information area will display. Click on the following options to affect the selected flight:

Speed Selection (underneath aircraft picture)

- **Minimum Speed** – The speed is shown in green
- **Cruise Speed** – The speed is shown in white
- **Military Speed** – The speed is shown in yellow
- **Maximum Speed** – The speed is shown in red

Altitude Selection (right side of aircraft picture)

- **Low** (LOW) **or Nap-of-the-Earth** (NOE) altitude
- **Cruise** altitude (CRU)
- **High** altitude (XHI), for certain aircraft

Ordnance Load

- **Selected Ordnance** (used for subsequent attack) highlighted in orange
- To select different ordnance, click on it (turns orange to show Selected)
- Ordnance prefaced by the symbol "*****" will be dropped if flight is attacked by enemy aircraft
- Ordnance prefaced by the symbol "**+**" is maximum possible for that configuration on that aircraft
- **Letters and Symbols** in parentheses next to the ordnance type provide clues to the type of ordnance that should be used based on the target selected (see AFEX Ordnance Reference Sheet on page 20).

Flight Identification (box above ordnance load):

- **Nationality Flag and Flying Unit Identity**
- **Number and Type of Aircraft in the Flight** – shown below the nationality flag and flying unit (NOTE: Aircraft icons may depict one or more aircraft of the same type packaged in the flight).
- **Command Organization** - usually the airbase shown below number and type aircraft

Endurance (how far the flight can fly before crashing from fuel starvation)

- **Fuel Status: Bingo** highlighted indicates that the flight is low on fuel and must return to its base, or refuel with a tanker.
- **Ordnance Status: Winchester** highlighted indicates that the flight has expended its primary mission ordnance, except possibly its cannon.
- **Refueling/Tanker** highlighted indicates that the flight is either currently being refueled or refueling other aircraft.
- **Range/Endurance** of the flight can be shown as a distance (range) or time (endurance). **White** indicates > 2/3 nominal range/endurance. **Yellow** indicates < 2/3 of its nominal range/endurance. **Red** (usually associated with Bingo) indicates < 1/3 of nominal range/endurance. Range/endurance can be extended by flying slower and /or higher.

Menu Descriptions

There are 9 contextual primary menus (structured menus that display associated options): the **Command Menu**, the **Status Menu**, the **View Menu**, the **Name Menu**, the **Highlight Menu**, the **Misc Menu**, the **Options Menu**, the **Settings Menu**, and the **Help Menu**. Clicking on any primary menu will display selectable options specific to that menu on one or two adjacent panels.

Command Menu

The Command Menu is used most often during gameplay. Therefore, this quick guide provides a detailed description of the selectable options under the Command Menu. Other menus are listed in the priority order of utilization in the main menu.



Drop Fuel Tanks drops any external fuel tanks from the selected flight.

Dispensing is not used in our exercise scenarios.

On/Off Passengers displays the **Passenger Dialog** so that passengers can be transferred between the current air base and the selected flight.

Load Cargo displays the **Cargo Dialog** so cargo can be loaded onto the selected flight.

Unload Cargo unloads all cargo from the selected flight.

Set Mission establishes the flight's mission type; primarily the functional color frame around the flight when the **Missions** option is selected.

Establish TOT displays the **Time-on-Target Dialog** so the flight's TOT (or time on station) can be established.

Change Load-Out changes the flight's weapons load-out as selected from presented load options to an available load-out. Weapon counts are reset to 0 and the flight is gradually re-armed. Full load of a weapon type for a configuration is represented with a plus (+).

Takeoff Time establishes the specific take off time of the selected flight.

Set Package causes the **Package Dialog** to display the selected flight's package or creates a new package with the selected flight as package leader. This option only applies to flights that are at an airbase or on the ground

Status Menu

Selectable options under the Status Menu provide information on the current operational effectiveness of airbases, key targets, missile sites, anti-aircraft artillery (AAA), radar sites, and command nodes. In addition, selecting fuel status and ordnance status displays the fuel and ordnance levels for each flight.



The View Menu

The main purpose of the View Menu is to allow the user to toggle map indicators on and off. These indicators include aircraft range, names on map, and aircraft orders.



A key selectable option is the Victory Condition which displays values earned by each side based on rescued crews, air refueling, and targets hit, as well As asset effectiveness. Only the host computer has access to this report.

Other Menus

The Name Menu, Highlight Menu, Misc Menu, Options Menu, Settings Menu, and Help Menu provide the user additional map display options as well as automatic settings the user may choose – several which may impact aircraft response such as Auto Fire and Auto Return.

Commanding Flights

Intuitive Tasking

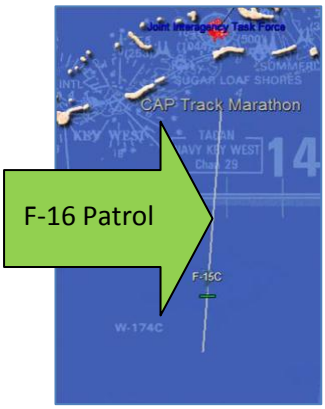
The TAV interface enables most flight actions to be accomplished simply by **Left**-clicking to select a flight and right clicking on its intended objective. Such actions include:

- Flying to a designated location
- Attacking an enemy flight
- Attacking an enemy target
- Escorting or following another flight
- Landing at a friendly base
- Rendezvous with a tanker

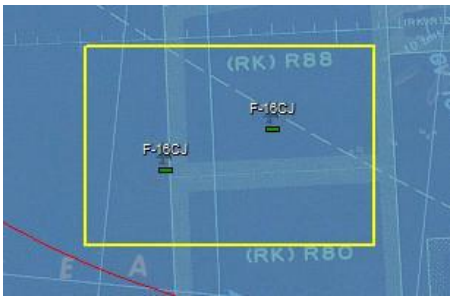
Compatible activity at click destinations is usually indicated by some type of symbology: red box for surface target, time for turnpoint/rendezvous, white circle for landing. If a more complex flight plan is needed, use **Shift-Right Click** to program turn points, altitude/speed changes, attacks, landings, etc.

Distances on the map can be measured by left-clicking on an origin point and dragging the cursor to the desired measurement point. Flights can be quickly returned to their last base by hitting the "R" key with the flight selected. Other intuitive TAV interface features are described below.

Patrol



A patrol area or "kill box" may be set for a flight by selecting a flight and using **Ctrl-Left Click** to drag a magenta-color box to the desired patrol size; the flight will be vectored to the patrol area and will randomly patrol within the box. Selecting a flight and right-clicking on the middle letter of a patrol track name sets that patrol track order for the flight—verify this by looking for times at both ends of the patrol track.



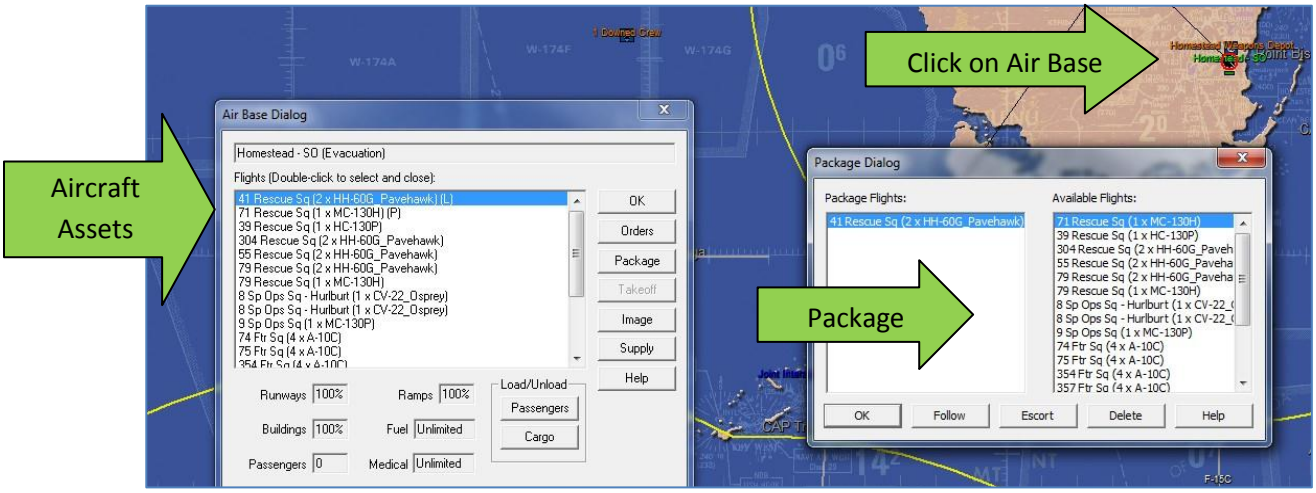
Multiple flights can be group-selected and given a common order by **Alt-Left Click** and dragging a yellow color box around the desired flights. Then right click on a designated position and selected flights will be highlighted in green and fly toward a common objective.

Firing on a Target

To manually fire on the target, **right-click** on the target while holding down the Control (**Ctrl**) key. When a flight is selected and it has a target, then the target flight is shown highlighted (in yellow). In most cases however, firing on enemy targets is performed automatically for you by the **Auto-Fire** option.

Force Packaging from an Airbase

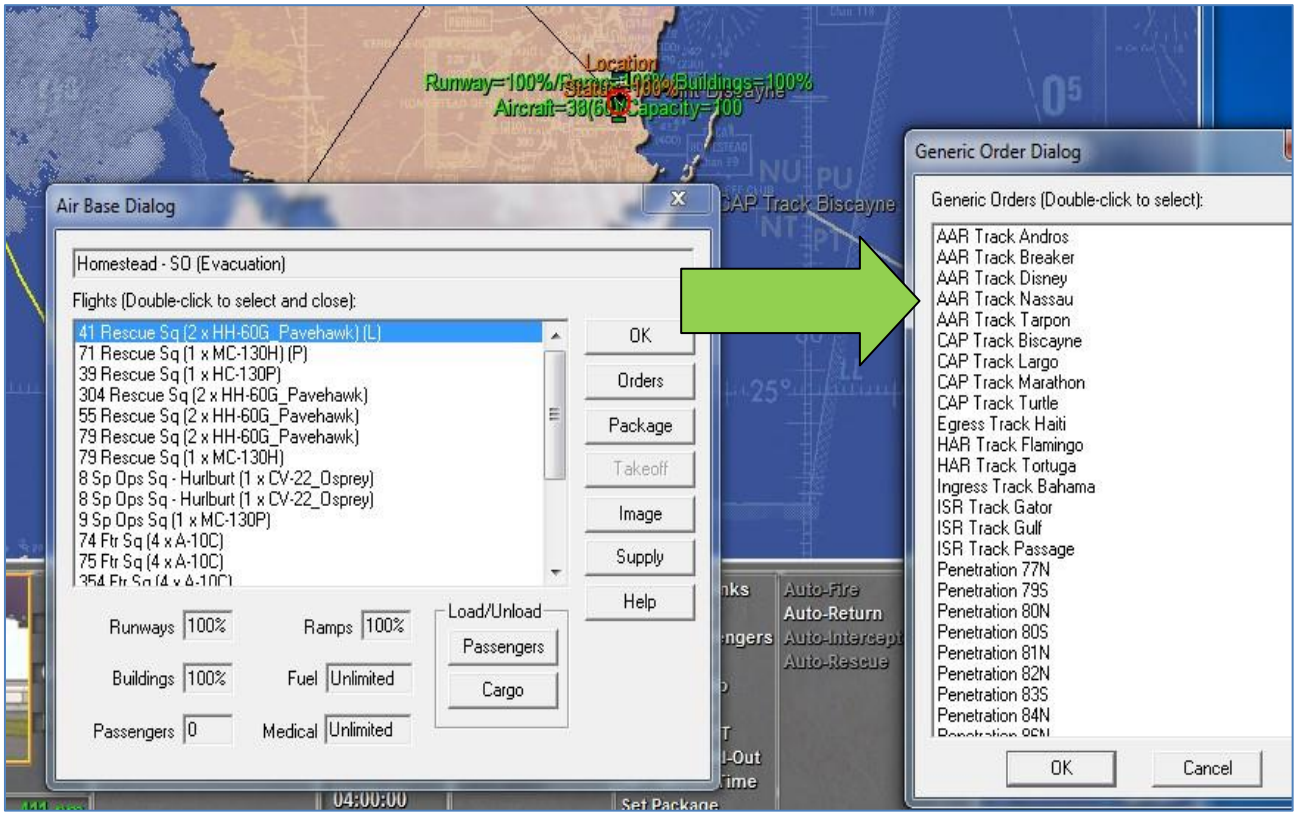
Effective force packaging is critical in AFEX. By double left clicking on an airbase, you can bring up the Air Base Dialog that lists aircraft currently parked at that base. Then you may force package a given



aircraft with other assigned aircraft and assign a mission. (See Appendix 3 on pages 30-32 for tips on Force Packaging.)

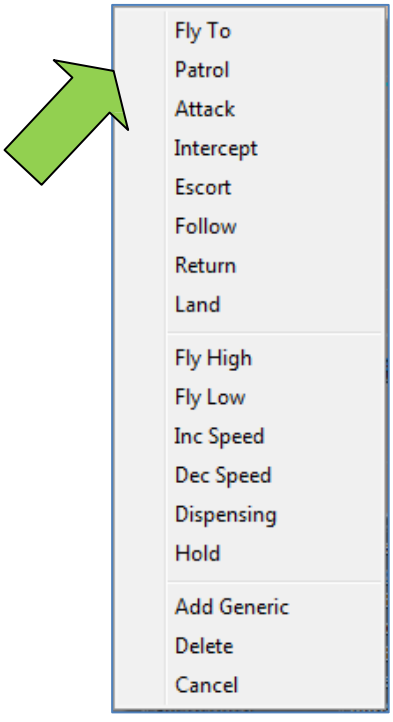
Assigning Orders from an Airbase

You may assign orders to a geographical point or track using the orders button.



Combining Orders

You can combine orders for a flight (Flight A) and issue more complex orders, by holding down the **Shift** key when you right click on the map. This will display the **Order Menu** and allow you perform additional functions:



Order Menu

Once the order menu appears, select the command and then right-click on the flight (Flight B) you wish to effect with your flight (Flight A).

Fly To – order to fly to the selected location will be added to the end of the orders for the selected flight. In this way, you can form a multi-segment flight path. Depending on the Segment setting in the Settings Menu, each segment will display the individual flight time or distance, or cumulative flight time or distance.

Patrol - order will be added to the orders of the selected flight to fly back and forth between the last location it has been ordered to or the current location if there are no previous orders and the selected location.

Attack - order to fly to a ground location and attack it will be added to the end of the orders for the selected flight.

Intercept - the selected flight (Flight A) will be given an order to intercept that flight (Flight B).

Escort - selected flight (Flight A) will be given an order to fly parallel with another flight (Flight B) and match its speed.

Follow - selected flight (Flight A) will follow another flight (Flight B) and duplicate its orders.

Return - order to return the selected flight to its home air base or ship and land will be added to the end of the orders for that flight.

Land - order to land the selected flight at the location will be added to the end of the orders for that flight.

Additional orders may be given to impact altitude, speed, etc.

Missions

Air-to-Ground Attack Missions

Force-Firing Weapons

By default, aircraft flights will only Auto-Fire one weapon at a time until that weapon impacts. Sometimes you may need to make your flight fire several missiles in succession. To force fire, use **Ctrl + right click** on the desired target(s). This is also useful to preemptively fire anti-radiation missiles on radar sites; most SAM sites will not go active with an ARM inbound, but if they do, the ARM will guide on the signal until impact. Force-firing active radar-guided air intercept missiles (AIM) like AIM-120 AMRAAM or MBD Meteor may be necessary when confronting multiple enemy aircraft flights within the weapon engagement zone.

Attacking SAM sites

A SAM site can either be **active** or **inactive**. When a site is active, anti-radiation missiles (ARM) or "R" munitions code weapons can guide on the SAM site's radar signal to destroy it. Other air-to-ground missiles can attack a SAM site regardless of its radar emission status. Destroying the radar renders the radar SAM battery useless, even if the site is not 100% destroyed. If total destruction is required, bombs (preferably area ordnance or "M" munitions code) are an optimum weapon, preferably after the SAM radar has been rendered inoperative.





SAM Status
When a SAM site is active, it is shown on the map in red. When it is inactive, it is shown on the map in gray.

SAM Radar
When a SAM is active, the radar associated with the site is being used. While this allows the SAM site to fire at enemy flights, it also allows the site to be identified under certain conditions and targeted by ARM (Anti-Radiation Missiles).

Number and Types of SAMs

If you turn on names on the map, either using the **Names on Map** option of the View menu or the **Names** toolbar button, then each SAM site will show the type of missile at that site. In the **Normal View**, you will also see how many missiles remain at that site.

Thermometer Display Option

If you turn on the thermometer display option, either from the **Show** submenu of the **Settings** menu or the thermometer toolbar buttons, then the top thermometer indicates the remaining missiles at the site and the lower thermometer indicates if the site still has a radar capability.

Spotting SAM Sites

By default, the locations of SAM and other missile sites are not known without ISR. There are situations that determine if a site is spotted.

Is Visible – The site is within the range of visibility of one of your flights. In this case, you will be able to spot the site regardless of other considerations.

On RWR – Certain aircraft carry Radar Warning Receivers (RWR). When a SAM site is active, then this receiver will allow the aircraft to spot the site. If the SAM site is not-active, then it cannot be picked up using RWR.

Spotting AAA Sites and Other Ground Locations

By default, AAA sites are unknown without ISR. If they fire, then their location becomes known and stays known for the duration of the scenario. Air bases are known locations and always spotted. Ground radar sites can be known by default or unknown depending on the scenario and site. If a radar site is unknown, then it can be detected using aircraft with ELINT capability.

Targeting Bases



Each air base consists of three components, which can be discretely targeted in an attack (designated mean point of impact or DMPI) to achieve a specific effect. Right-clicking on an enemy airbase with a flight of aircraft selected brings up the Air Base Targeting Dialog requesting clarification of the attack DMPI. Planning an Attack on an airbase from the Flight Planning Menu (Shift-Rt Click) also brings up the Air Base Targeting Dialog.

Runways –Runways are used for aircraft takeoffs and landings, however airbases have other hard surfaces (taxiways, etc.) that can be used for takeoffs and landings. Runway damage will delay (but not prevent) enemy takeoffs and landings; some takeoffs and landings still occur even with total runway destruction. Airbase damage repairs over time with Construction units present.

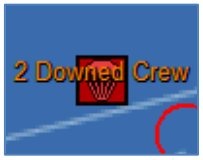
Ramps – Ramps are where aircraft are parked on the airbase. Ramp damage permanently destroys a proportional percentage of parked aircraft.

Buildings – Buildings house fuel and ordnance for refueling and reloading. Building damage permanently destroys a proportional percentage of stored fuel and ordnance.

Search and Recovery Missions



Helicopters/Ospreys can be used to rescue downed crew members from crashed flights over water or over enemy areas. To rescue a friendly (blue) Downed Crew, select the rescue flight you plan to use then right-click on the icon depicting the flight you wish to rescue. This orders the flight to fly to that location, land, and pick up the downed crew. A successful rescue is indicated by "Has Rescue" legend illuminated on the rescue aircraft and "Crash Site" appears at the spot on the map where the downed crew was rescued. Helicopters and Ospreys can rescue multiple downed crews up to the load Capacity limit of the aircraft. The typical weight factor for one downed crewman is around 200 lbs (load capacity for an Osprey is 10,000 lb or about 50 downed personnel).

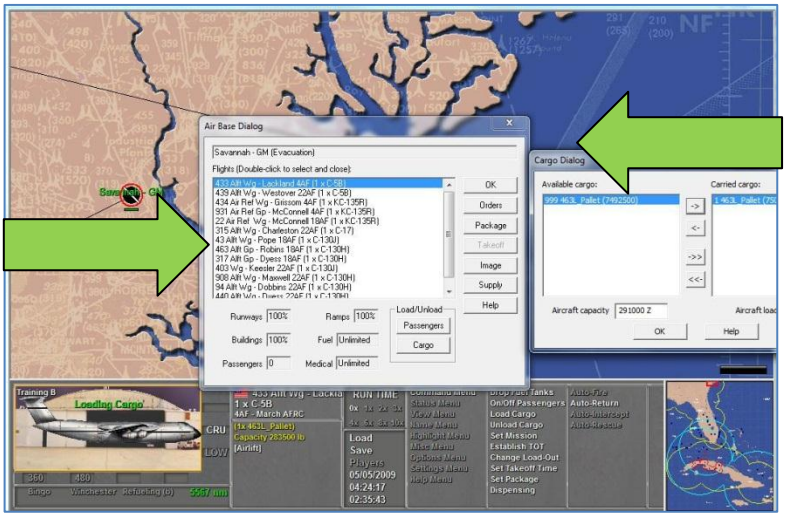


Enemy downed crew appears as a red downed crew icon and cannot be picked up in this scenario. Friendly downed crew automatically unload at any airfield or ship where the helicopter lands. It is important to protect downed crew from enemy capture. Provide RESCAP (rescue close air support) by sending fighter/attack aircraft with strafe capability over the downed unit to delay capture by enemy forces. Otherwise you may see the following message:



Cargo Airlift Missions

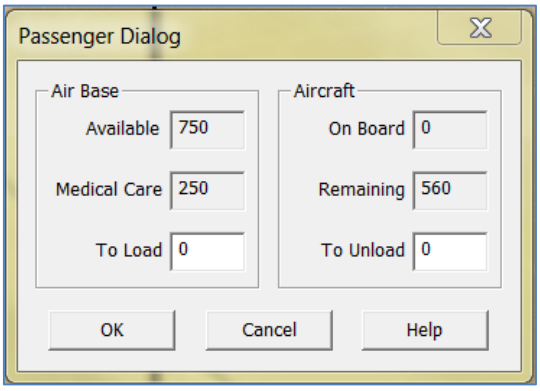
Load cargo by first selecting the base where the cargo aircraft is located. To see cargo available at the base, select the "Cargo" button under "Load/Unload". Transfer cargo from the base to the selected aircraft by using the right arrows. Using the double-arrow loads maximum quantity of the cargo type selected at once. If the destination is an air base, follow the same procedures as above to offload the cargo. If the destination is an austere location (for example where SOF are located), land or airdrop cargo by using either the "Load Cargo" or "Unload Cargo" menu option (right click on the aircraft picture if you can't see the full menu).



To ensure the unit received the supplies, select "Target Status" from the menu items, highlight the unit, then click the "Supply" button. Cargo aircraft may be unable to land if the base is full (exceeds max on ground or MOG) or if takeoffs are scheduled at the base (departing aircraft on runway); some airbases may be damaged with reduced capacity. If you can't land, try having some of your empty aircraft take-off from the base.

Civilian Airlift Missions

Civilians can either be evacuated from airbases as Passengers or picked up from assembly areas in friendly territory by helicopters/Ospreys like Downed Crew. P assengers are loaded onto dedicated transports (*cargo and passengers can't be mixed on the same aircraft*) using procedures similar to Cargo above, using a specific **On/Off Passengers** menu option. To pick up passengers at an airbase, type in the desired number in the **To Load** window, up to the capacity number indicated in the **Remaining** window. To unload, repeat the process by typing in the **To Unload** window.



Refueling Missions



Refueling is critical to prevent unnecessary friendly losses from fuel kills. Any aircraft that runs out of fuel, or does not have enough fuel to return to base when the game ends, will be listed as a fuel kill. Refuel your aircraft by first matching the aircraft refueling type (**boom (b)** or **probe (p)**) and possible airspeeds (slow for helicopters) with a compatible tanker. Tankers should be patrolling at minimum airspeed to facilitate rendezvous with overtaking receiver aircraft. Next, select the flight that requires fueling and **Right Click** on the tanker. You will see "**Refueling**" highlighted in the flight information area of the receiver aircraft ("**Tanker**" for the tanker when transferring fuel) when the connection is successful and fuel is being transferred. Fuel kills can also be minimized by avoiding the use of Maximum speed settings—use Max Power (red speeds) only in emergencies and only for the minimum time needed. Keep a watchful eye on fighter aircraft (particularly stealth) with limited fuel and high burn rates.

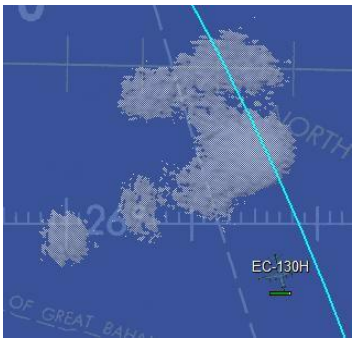
Battle Damage Assessment (BDA) Missions

Satellites may be scheduled to over-fly the area. You have no control over this satellite but it will provide you with intelligence about enemy locations as it flies over them. Satellites will reveal many enemy units as they pass over, however, only static targets will remain visible. Mobile units will disappear once the satellite can no longer see them.

Static targets that have been destroyed will be displayed on the map as a flaming target box (no white symbology). Mobile targets, such as missile launchers, will be displayed on the map as a flaming icon once destroyed. Note that although you may have crippled a SAM site by taking out its radar (noted by a white (not red) weapon trace below the SAM site), the flaming icon will only appear if the SAM has been 100% destroyed. Try to watch the orange messages flash at the top of the screen, particularly when "(radar out)" is noted.

Variables

Weather



Certain scenarios will begin with weather conditions shown by clouds. These represent low-altitude clouds and areas of reduced surface visibility. They will have an impact on the ability to spot certain aircraft and ground locations.

- When a ground location is under cloud cover, it can't visually see aircraft but can still see them with radar.
- When a ground location is under cloud cover, it can't be attacked except by aircraft with bombing radar or with ordnance that uses GPS guidance.
- Recon cannot be conducted against a ground target under cloud cover.
- Low ceiling aircraft, helicopters, and other aircraft that are flying low are considered to be low enough that they fly through clouds, instead of above them as an aircraft at normal cruise altitude would.
- When aircraft fly though clouds, they have no ability to see the ground or other aircraft. They can detect other aircraft with radar but not visually or using IR.
- Aircraft flying through clouds cannot fire or be fired upon by IR Homing air-to-air missiles or missiles fired visually from the ground.

Night

Depending on the visibility, conditions in a scenario are considered to be at **night**. This is shown in the upper-left hand corner of the Main Chart. The visibility value that determines night is Parameter Data and by default is 1 statue mile. When night conditions exist, then it is not possible to make a bombing strike against a ground target unless the attacking aircraft has FLIR, bombing radar, or the ordnance has GPS guidance.

Launch Intervals

Associated with each side are parameter data values called **Launch Intervals**. These determine how long it takes after a flight takes off or lands at an airbase before the next flight can take off or land. Each time a flight takes off or lands, this timer is set equal to a calculated delay.

Landing

When a flight attempts to land at an air base, the status of the air base affects their ability to do that. For example, there must be sufficient parking space for the landing flight, as given by the ramp status times the nominal capacity of the air base, before a flight can land.

Flight Collisions

When a flight of more than one aircraft is flying low in weather, there may be a chance of a collision between aircraft in the flight. The probability increases as the number of aircraft in the flight increases. This effect applies to all helicopters as well as airplanes flying at low altitude. The probability doubles at night.

Specialized Features

Recon Aircraft



Recon aircraft are identified in the Database Dialog or by the words "Has Recon" in the Flight Information area. Recon aircraft can be used to "attack" a ground site. When the overflight occurs, the Recon aircraft performs a recon of the site and the words "Has Recon" are highlighted.

Once a recon aircraft has performed a recon of a site, any damage associated with that site at the time is counted towards Recon Points in the determination of victory conditions. In addition, any subsequent damage of that location is given a bonus towards the victory point calculation. The magnitude of these additional points is determined by the Pre-Recon and Post-Recon parameter data values.

When a recon aircraft has performed a recon of an air base, then an air base recon entry is created that shows the information associated with that recon. These air base recon entries can be viewed by using the **Air Base Recon** option of the **Status** menu. A recon aircraft can only perform one recon mission during a scenario and a given ground location can only have one recon performed against it in a scenario.

Radar Jamming Aircraft



Certain aircraft have an inherent radar jamming effect defined in the aircraft database. This jamming ability is determined by a range for the radar jamming effect and an angle measured from the wing tips of the jamming aircraft. The coverage area is shown on the map using **bright purple** wedges. Radar sources that are within the range and angle defined by the jamming data are affected. This effect is displayed on the map using **bright green** wedges. When an aircraft is inside this wedge, it cannot be detected by the affected radar.



Radar sources that are within the range and angle defined by the jamming data are affected. This effect is displayed on the map using **bright green** wedges. When an aircraft is inside this wedge, it cannot be detected by the affected radar.

Jamming Pods

Certain aircraft can carry a Stand-Off-Jamming (SOJ) pod. This provides the aircraft with the ability to perform radar jamming just like a standard radar jamming aircraft. The range and angle associated with the pod in the ordnance database determines its jamming capability.

Airborne Ground Radar

Certain aircraft carry radar that is capable of detecting ground targets. This ability is identified in the aircraft database in terms of distance and angle. The angle of coverage is measured from the wing tips of the aircraft. This area is displayed on the map using **dark green** wedges. If a ground Target that is otherwise unknown is in the range and coverage of the ground radar carried by the aircraft, then it becomes known to the detecting side.



Communication Jamming

Certain aircraft have the ability to perform COMJAM against enemy command nodes. This is shown in the aircraft information area. Their effective range is shown on the map using a purple jamming ring. When an enemy command node comes within jamming range of the aircraft, then a purple line is drawn to that node. Their disruptive effect is proportional to their distance to the command node relative to their maximum jamming range and thus varies from 0% at maximum range up to 100% when they are directly overhead.

Electronic Counter Measures (ECM) Effects

Certain aircraft have an on-board self-protection ECM value given by the aircraft database. All aircraft may also carry ECM pods defined in the ordnance database. This feature gives the aircraft increased protection from weapons that use radar tracking. ECM effects apply to radar-guided SAMs fired at the aircraft as well as fire from radar-directed AAA guns.



ELINT

Certain aircraft are capable of Electronic Intelligence (ELINT). This is indicated in the alternative unit display and the range of the capability is shown. When an enemy aircraft is within this range and it has been spotted on radar, then the flight can be selected on the map and the type of aircraft in the flight is known.



AFEX Ordnance Reference Sheet

These symbols may be found for each aircraft in the Flight Information Area next to selected ordnance.

(*A) **Active-Radar Tracking**– These missiles have their own tracking radar. Once the missile is fired, the firing aircraft is free to maneuver with no danger of the missile losing tracking on the target aircraft.

(^B) **Anti-Ballistic** - An Anti-Ballistic Missile (ABM) weapon fired at ballistic Surface-to-Surface missiles.

(C) **Cruise Missile**– This ordnance is a low-flying missile.

(E) **Laser**– This represents laser weapons which fire at enemy Surface-to-Surface missiles. They do not require tracking, only that the target is within the weapon envelop before firing. They cannot be used against enemy aircraft and they cannot be fired through clouds.

(^F) **Fixed Only** - The weapon can only be fired against fixed, non-moving targets.

(*G) **GPS Guidance**– This ordnance is not affected by weather and visibility limitations as ordnance guided visually would have. However, it can be disrupted by GPS jammers associated with the target.

(^H) **Can Fire High**– Allows the weapon to be fired at high-flying aircraft.

(^L) **Low Altitude** - The weapon is a low-altitude SAM that can only reach low flying aircraft.

(M) **Submunition**– This ordnance can be used effectively against Area Targets while standard ordinance has little effect.

(N) **Runway Penetrating**– This ordnance when used against Air Bases causes 10 times the normal damage it would cause given its explosive rating.

(*O) **Manual Tracking**– These missiles or bombs are guided to the target by the crew of the attacking aircraft. They must be released within visual range of the target and the attacking aircraft must remain in the visual range of the target until the ordnance hits.

(*P) **Electro-Optical Tracking**– This ordnance has a camera that allows it to lock-onto the target image and then to guide itself to the target after release. This ordnance must be released within visual range of the target, but once released, the attacking aircraft is free to maneuver.

(*R) **Radar Homing Tracking**– This ordnance can be fired at an active-radar SAM site, a radar site, an air base that has radar, a AAA site that uses radar, or at a Target or missile site having AAA that uses radar.

(*V) **Visual Tracking**– An example would be rockets, which are fired at a distance, but have no tracking and so are unguided once fired. This ordnance must be fired or dropped within visual range of the target.

(*R) **Radar-Homing Tracking** - These missiles can be fired at an enemy flight that has the firing flight in its radar fan but if the enemy flight turns away, then the missile losses its tracking.

(*S) **Semi-Active Tracking**– These missiles use a radar tracking signal generated by the firing aircraft. Once fired they require the firing aircraft continue to track the target to continue to home on the reflected radar signal.

(T) **Penetrating**– This ordnance has effect against Hard Targets whereas non-penetrating ordnance has very little effect against them.

AFEX Keyboard (Commands) Reference Sheet

Mouse Commands

Left-Click: Select flight or SAM site.
Right-Click: Fly to that location, intercept enemy flight, escort friendly flight, or attack enemy ground target.
Alt Right-Click: Cause selected flight to patrol between current location and specified location.
Ctrl Right-Click: Force fire selected ordnance at enemy flight or ground location, or fire SAM at enemy flight.
Shift Right-Click: Display Order Menu so that orders can be combined for selected flight.
Double Left-Click: On airbase, display Air Base Dialog which shows flights currently assigned to that air base.

Standard View Commands

1, 2, 3, 4: View main map at specified zoom level.
Left-Click Jump Map: Jump main map to specified location.
Mouse at Screen Edge: Scroll main map in specified direction.
Hotkey G: Show Generic Orders and select a desired track or point for that flight.
Hotkey R: Return to home Airbase and Land.

Selected Menu Commands

Missions: Toggle color-coded highlight indicating the assigned mission for each flight
Mission Log: View a log of the notices that have appeared at the top of the screen.
Victory Condition: View a summary of friendly and enemy losses, as well as other point values.
Prtzd Target List: View estimated damage to be inflicted on targets based on flights attacking those targets.
Load/Unload Cargo: Use these menu commands to load/unload cargo on a cargo plane.
Auto-Fire: Automatically fire ordnance at enemy flights.
Auto-Intercept: Automatically intercept enemy flights within range.
Auto-Return: Automatically return to base when low on fuel.
Status Values: Toggle % effectiveness of ground targets.
Misc. Status Selections (e.g. Air Base Status): View the % effectiveness of various targets around the map.

Specialized Range Indicators

Yellow Area: Range of on-board or ground radar.
Red Area: Range of selected missile or surface-to-air threat ordnance (SAM or AAA).
Blue Circle: The current visibility range.
Black Circle: Flight range of aircraft.
Brown Circle: Distance to horizon for Low Flying aircraft.
Purple Area: Area of Jamming broadcast. **Bright**
Blue Area: Area of SIGINT coverage. **Bright Green**
Area: Area of effective Jamming. **Bright Green**
Line: Direction of ELINT detection.
Dark Green Area: Area of Ground Radar coverage.

Flight Plan Commands (Mini-Menu Shift Rt-Click)

Fly To: Use this command when plotting out multiple waypoints for a flight to fly. Otherwise just Right-Click.
Patrol: Use this command when you wish to set up a CAP or orbit between two points. Same as Alt Rt-Click.
Attack: Use only to attack a specific area of a base (e.g. ramps). Otherwise Right-Click to attack entire base.
Hold: Use to enter a time in minutes and seconds to hold at that location. Displayed as a white diamond.
Delete: The last order for the selected flight will be deleted.
Cancel: No order is given.

Appendix 1:

OPERATION AFEX
EXECUTION ORDER (EXORD)
COMBINED FORCES COMMAND OPLAN

STRATEGIC REGIONAL SITUATION:

INTERCEPTS OF SECRET TALKS BETWEEN CUBA, VENEZUELA, AND RUSSIA HAVE REVEALED THE EXISTENCE OF AN ALLIANCE AIMED AT DECREASING U.S. INFLUENCE IN AND AROUND CENTRAL AND SOUTH AMERICA AND INCREASING RUSSIA'S FOOTHOLD IN THE WESTERN HEMISPHERE.

IN RECENT YEARS, VENEZUELA AND RUSSIA HAVE ASSISTED CUBA WITH REBUILDING THEIR CAMILO CIENFUEGOS REFINERY, EXPANDING THEIR ABILITY TO REFINE CRUDE OIL. ALTHOUGH EXPLORATORY ENDEAVORS HAVE ENCOUNTERED GEOLOGICAL AND OTHER OBSTACLES, RUSSIA CONTINUES TO POUR MONEY INTO CUBA'S COFFERS. IN RETURN, CUBA HAS AGREED TO SEVERAL AS YET UNCONFIRMED STIPULATIONS WITH VENEZUELA AND RUSSIA.

ANALYSIS OF THE DATA INDICATES THAT CUBA WILL ALLOW VENEZUELA JOINT EXPLORATORY RIGHTS ALONG WITH OTHER INTERESTED COUNTRIES SUCH AS INDIA AND CHINA WHO HAVE ALREADY INVESTED OVER 1 BILLION DOLLARS IN THE ENTERPRISE. THE U.S. GEOLOGICAL SURVEY HAS ESTIMATED THAT CUBA'S OFFSHORE WATERS HAVE "SIGNIFICANT UNDISCOVERED CONVENTIONAL OIL POTENTIAL - BETWEEN 4.6 BILLION AND 9.3 BILLION BARRELS. CUBAN OFFICIALS ESTIMATE THE POTENTIAL RESERVES AT 20 BILLION.

IN ADDITION, CURRENT DATA POINTS TO AN AGREEMENT WITH RUSSIA FOR THE DEVELOPMENT OF AN UNDERGROUND BASE IN RETURN FOR FURTHER EXPERTISE AND EQUIPMENT FOR OIL EXPLORATION AND PRODUCTION. UNCONFIRMED HUMAN INTELLIGENCE HAS ALSO SUGGESTED THAT VENEZUELA IS OFFERING A JOINT MILITARY SUPPORT AGREEMENT FOR THE DELIVERY OF TWO DOZEN SUKHOIFIGHTER JETS, MODEL SU-30MK FROM RUSSIA. THERE IS ALSO SOME INTELLIGENCE THAT INDICATES AS PART OF THE UNDERGROUND AIRBASE DEAL WITH CUBA, RUSSIA WILL PROVIDE CUBA SEVERAL SU-30S AS WELL AS PILOTS AND OR TRAINING.

ANOTHER CONCERNING POSSIBILITY EXISTS THAT RUSSIA COULD HAVE COVERTLY DELIVERED A GREATER MILITARY CAPABILITY TO VENEZUELA, WHICH IN TURN COULD EASILY BE FERRIED TO CUBA AT A MOMENT'S NOTICE. IF RUSSIA HAS ALSO PROVIDED EXPERIENCED PILOTS OUR THREAT BECOMES GRAVE SHOULD ANY OVERT MILITARY ACTION TAKE PLACE.

ON THE DOMESTIC FRONT, VENEZUELA RECENTLY HOSTED SEVENTH ANNUAL CONTINENTAL CONFERENCE IN SOLIDARITY WITH CUBA. AT THIS GATHERING, THE NEWLY ELECTED PRESIDENT OF VENEZUELA SOUGHT TO STRENGTHEN HIS POSITION OF POWER AND INFLUENCE IN THE REGION WHILE THE PRESIDENT OF

CUBA SOUGHT TO REDEFINE HIS LEGACY IN LIGHT OF THE RAPIDLY APPROACHING EXPIRATION OF HIS FAMILY'S RULE.

MEANWHILE, THE U.S. HAS BEEN EMBARRASSED BY RECENT HIGH-LEVEL INTERNAL SECURITY LEAKS THAT HAVE REVEALED SECRET DATA COLLECTION INSTRUMENTS AND TACTICS. THIS HAS FORCED THE U.S. TO ADDRESS ALLEGATIONS OF SPYING TO PARTNER NATIONS IN THE REGION AND HAS PROMPTED SOME NATIONS TO BECOME MORE SYMPATHETIC TO POSITIONS ADVOCATED BY CUBA AND VENEZUELA.

SITUATION UPDATE:

THE U.S. CURRENTLY HAS 600 NONCOMBATANTS STATIONED AT GUANTANAMO BAY NAVAL BASE. CUBA HAS THREATENED ATTACK ON GUANTANAMO BAY NAVAL BASE IF THE U.S. DOES NOT EVACUATE WITHIN 10 DAYS. YOU ARE DIRECTED BY THE SECRETARY OF DEFENSE TO PLAN FOR THE POSSIBLE EVACUATION OF GUANTANAMO BAY NAVAL BASE, AS WELL AS A MILITARY RESPONSE TO ANY HOSTILITIES.

AIR TASK FORCE COMMANDER MISSION STATEMENT:

THE AIR TASK FORCE WILL CONDUCT AIR OPERATIONS TO SUPPORT THE AERIAL EVACUATION AND RETURN TO CONUS OF NONCOMBATANTS CURRENTLY STATIONED AT GUANTANAMO AIR BASE CUBA.

ASSUMPTIONS:

1. CHEMICAL AND NUCLEAR WEAPONS USE ARE RESTRICTED.

2. CUBA MAY CONSIDER US AIRCRAFT ENTERING THEIR AIRSPACE AS PROVOCATION AND REASON TO ATTACK.

3. OFFENSIVE OPERATIONS DESIGNED TO NEUTRALIZE CUBAN DEFENSIVE OPERATIONS MAY BE NECESSARY TO ENSURE NONCOMBATANT EVACUATION.

AIR TASK FORCE TASKS:

1. DETER CUBAN AGGRESSION.

2. POSTURE ASSETS IN PREPARATION FOR A NONCOMBATANT EVACUATION. IF

DETERRENCE FAILS, SPECIFIED FUNCTIONAL TASKS ARE:

1. GAIN AND MAINTAIN AIR SUPERIORITY WITHIN THEATER OF OPERATIONS.

2. CONDUCT AND SUPPORT COUNTEROFFENSIVE OPERATIONS.

3. DEGRADE ENEMY CONVENTIONAL MILITARY POWER.

4. PREVENT AIR OR MISSILE ATTACKS.

5. CONDUCT NEO EVACUATIONS.

6. RESUPPLY FORWARD OPERATING BASES (FOB).

7. COORDINATE AND ASSIST IN THE RECOVERY OF DOWNED AIRCREWS.

Appendix 2:
Team Descriptions, Assigned Aircraft and Tasks



TEAM 1: GLOBAL PRECISION ATTACK

This team is responsible for identifying and neutralizing or destroying key targets whose elimination will contribute to the overall accomplishment of air campaign objectives. Global Precision Attack is identified as a function of the Air Force in DoDD 5100.01, to include strategic attack, interdiction, close air support, and prompt global strike. Team tasks are aligned primarily with the doctrine of Strategic Attack. However, the team may also support other types of Air Force operations listed under USAF Doctrine Volume 4 Operations.

STRATEGIC ATTACK – Strategic attack involves the systematic application of force against enemy systems and their centers of gravity, thereby producing the greatest effect for the least cost in lives, resources, and time. Vital systems affected may include leadership, critical processes, popular will and perception, and fielded forces.

TEAM TASKS:

1. UTILIZE STRATEGIC BOMBER ASSETS TO WEAKEN OR ELIMINATE ADVERSARY's ABILITY TO ENGAGE.

2. SUPPRESS, NEUTRALIZE, OR DESTROY ENEMY SURFACE-TO-AIR MISSILE NETWORK.

3. INTERDICT AND ATTACK GROUND FORCE UNITS.

Available organizations for supporting Strategic Attack are assigned to **Moody AFB**. At the start of game play, several of these assets are already patrolling the air. The host computer identifies these assets under **Strategic Attack**. These include:

B-1B Bomber (Lancer)
- BSU-49: Effective on all targets, recommend runways and control centers

B-2 Bomber (Spirit)
- BSU-50: Effective on all targets, recommend runways and control centers

B-52H Bomber (Stratofortress)
- AGM-86C Conventional Air Launched Cruise Missile (CALCM): Effective on radar sites and other point targets
- Aerial Radar Jammer

VICTORY CONDITION SCORING:

This team will need to develop SMART objectives based on overall team tasks in support of the overall air campaign plan. Effectiveness will be assessed partly on their effectiveness in destroying or neutralizing prioritized targets.



TEAM 2: COUNTERAIR (OFFENSIVE)

This team is responsible for achieving and maintaining air superiority. Offensive operations including operations required to gain and maintain air superiority are listed in DoDD 5100.01 as a function of the Air Force. Team tasks are aligned primarily with the doctrine of Counterair Operations. However, the team may also support other types of Air Force operations listed under USAF Doctrine Volume 4 Operations.

COUNTERAIR - Counterair operations are conducted across all domains and determine the level or degree of air control. Air control describes a level of influence in the air domain relative to that of an adversary, and is categorized as parity, superiority, or supremacy.

TEAM TASKS:

1. ACHIEVE AND MAINTAIN AIR SUPERIORITY OVER AREA OF OPERATION IN SUPPORT OF THEATER OBJECTIVES.

2. SWEEP AGGRESSIVELY, ENGAGE, AND DESTROY ANY AIRCRAFT THREATENING US AIR OPERATIONS.

3. DESTROY ENEMY AIRCRAFT ON THE GROUND.

4. NEUTRALIZE ENEMY SURFACE-TO-AIR MISSILE NETWORK USING SEAD ASSETS.

5. PROVIDE ESCORT TO ASSETS SERVING IN AIRLIFT, AIR REFUELING AND CSAR ROLES.

Available organizations for supporting Counterair are assigned to **Patrick AFB**. At the start of the game, several of these assets are already patrolling the air. The host computer identifies these assets under **Counterair-Offensive**. These include:

F-22 Fighter (Raptor)
- AIM-120 Air to Air Missile, Medium range, multiple aspect
- AIM-9 Air to Air Missile, Short range, Heat seeking cannot fire nose to nose

F-15C Fighter (Eagle)
- AIM-120 Air to Air Missile, Medium range, multiple aspect
- AIM-9 Air to Air Missile, Short range, Heat seeking cannot fire nose to nose

F-16CM Fighter (Fighting Falcon)
- AIM-120 Air to Air Missile, Medium range, multiple aspect
- AIM-9 Air to Air Missile, Short range, Heat seeking cannot fire nose to nose

VICTORY CONDITION SCORING:

Team will need to develop SMART objectives based on overall team tasks in support of the overall air campaign plan. Team's effectiveness will be assessed partly on their effectiveness in destroying or neutralizing SAM targets. In addition, they will be graded on their Air-to-Air and Air-to-Surface aircraft kills.



TEAM 3: COUNTERAIR (DEFENSIVE)

This team is responsible for defending against enemy air threats. In addition, they should assist with other counterair missions such as sweep and escort. Offensive operations including operations required to gain and maintain air superiority are listed in DoDD 5100.01 as a function of the Air Force. Team tasks are aligned primarily with the doctrine of Counterair Operations. However, the team may also support other types of Air Force operations listed under USAF Doctrine Volume 4 Operations.

COUNTERAIR - Counterair operations are conducted across all domains and determine the level or degree of air control. Air control describes a level of influence in the air domain relative to that of an adversary, and is categorized as parity, superiority, or supremacy.

TEAM TASKS:

1. MAINTAIN LOCAL AIR SUPERIORITY.

2. ASSIST OCA ASSETS WITH SWEEP AND ESCORT FUNCTIONS.

Available organizations for supporting team tasks are assigned to **MacDill AFB**. At the start of game play, several of these assets are already patrolling the air. The host computer identifies these assets under **Counterair - Defensive**. These include:

F-15E Fighter (Strike Eagle)
- AGM154A- Joint standoff weapon (JSOW) good for radar and SAM sites
- GBU-31-Joint direct attack munition (JDAM) Good against SAMs and Control Centers, Marginal against Ramps, Buildings, and Runways about 28% damage per attack

VICTORY CONDITION SCORING:

The team will need to develop SMART objectives based on overall team tasks in support of the overall air campaign plan. Team effectiveness will be assessed partly on their effectiveness in deterring friendly air-to-air and air-to-surface losses. In addition, they will be assessed on their ability to minimize aircraft lost due to loss of fuel (requires coordination with Team 4, Air Mobility).



TEAM 4: AIR MOBILITY

This team is responsible for refueling and moving passengers and cargo. Team tasks are primarily aligned with the Air Force doctrine of Air Mobility. However, the team may also support other types of Air Force operations listed under USAF Doctrine Volume 4 Operations.

AIR MOBILITY - Joint doctrine defines air mobility as "the rapid movement of personnel, materiel, and forces to and from or within a theater by air." DoDD 5100.01 identifies a function of the Air Force as providing global mobility to employ and sustain organic air and space forces and other Military Service and US Special Operations Command forces, as directed, to include airlift forces for airborne operations, air logistical support, tanker forces for in-flight refueling, and assets for aeromedical evacuation.

TEAM TASKS:

1. AIRLIFT MEDICAL SUPPLIES TO DESTINATION SITE OF NONCOMBATANT EVACUATION OPERATION.

2. AIRLIFT AMERICANS STATIONED AT GUANTANAMO AIR BASE TO DESTINATION SITE OF NONCOMBATANT EVACUATION OPERATION.

3. PROVIDE AIR REFUELING TO THEATER ASSETS.

Available organizations for supporting Air Mobility are assigned to **Savannah AFB**. At the start of game play, several of these assets are already patrolling the air. The host computer identifies these assets under **Air Mobility**. These include:

C-17, C-5, and C-130E/J – Cargo (ability to carry passengers consistent with Air Force fact sheets)

KC-135 and KC-10 – Air Refueling (ability to transport fuel consistent with Air Force fact sheets)

VICTORY CONDITION SCORING:

The team will need to develop SMART objectives based on overall team tasks in support of the overall air campaign plan. Team's primary role is to move passengers from Guantanamo to CONUS. In addition, the team's success will be measured by their resupply of air bases and refueling effectiveness.



TEAM 5: GLOBAL INTEGRATED ISR (GIISR)

This team is responsible for capturing and disseminating an accurate picture of enemy assets. In addition, the team may be tasked to assist counterair teams with SEAD and Jamming. The team may also support other types of Air Force operations listed under USAF Doctrine Volume 4 Operations.

GLOBAL INTEGRATED ISR - The Air Force defines GIISR as "cross-domain synchronization and integration of the planning and operation of ISR assets; sensors; processing, exploitation and dissemination systems; and, analysis and production capabilities across the globe to enable current and future operations." DoDD 5100.01 lists as an Air Force function the requirement to provide timely, global integrated ISR capability and capacity from forward deployed locations and globally distributed centers to support world-wide operations.

TEAM TASKS:

1. DETECT TAKEOFFS AND CLASSIFY ALL ENEMY AIR MISSIONS, MONITORING THEIR AIRBORNE FLIGHT PATHS FOR ENGAGEMENT IF THEY POSE A THREAT TO COALITION AIR OPERATIONS.

2. DETECT AND CLASSIFY ENEMY'S EARLY WARNING RADARS, SAM SITES, AND COMMAND NODES.

3. SUPPRESS AND/OR JAM ENEMY AIR DEFENSES.

Available organizations for supporting GIISR are assigned to **Tyndall AFB**. At the start of game play, several of these assets are already patrolling the air. The host computer identifies these assets under **GIISR**. These include:

RQ-4 ISR (Global Hawk) & MQ-9 ISR (Reaper) – Imagery and Multi Spectrum Signals Intelligence

E-3 ISR (Sentry) - Airborne Early Warning Control (AWACS)

U-2S ISR (Dragon Lady) - Imagery and Multi Spectrum Signals Intelligence

RC-135V ISR (Rivet Joint) - On board sensors detect, identify and geo-locate signals, works with the E-3 to provide air picture

F-16CM (Fighting Falcon) Iron Hand, Wild Weasel
- Suppression of Enemy Air Defenses (SEAD) are military actions to suppress enemy surface-based air defenses Surface-to-air missiles (SAMs) and anti-aircraft artillery
- AGM-88 HARM High Speed Anti-radiation missiles, excellent against SAMs and Radar, good range
- AGM-65K good against SAMs and Radar, poor range

EA-6B ISR Jammer (Prowler) - aerial radar jamming

EC-130H ISR (Compass Call) - Cyber Attack, Jamming

VICTORY CONDITION SCORING: Team will be assessed primarily on their ability to detect and classify air and ground enemy capabilities.



TEAM 6: PERSONNEL RECOVERY

This team is responsible for recovering downed aircrews and returning them safely to base. In addition, they may be tasked to provide aircraft for close air support and specialized refueling. However, their primary tasks are aligned with the Air Force function of Personnel Recovery.

PERSONNEL RECOVERY OPERATIONS (PRO) - The primary mission of Air Force PRO is to use a combination of specially trained Airmen and unique equipment to recover any isolated personnel. DoDD 5100.01 identifies global personnel recovery operations as a function of the Air Force. Although traditionally PRO assets have focused on the recovery of downed aircrews, recent experiences suggest that Air Force PRO forces are responsible for the recovery of any isolated personnel.

TEAM TASKS:

1. RECOVER DOWNED AIRCREWS

2. RETURN CREWS SAFELY TO BASE.

3. PROVIDE CLOSE AIR SUPPORT. (Global Precision Attack function)

4. PROVIDE SPECIALIZED REFUELING. (Special Operations function)

Available organizations for supporting Personnel Recovery are assigned to **Homestead AFB**. At the start of game play, several of these assets are already patrolling the air. The host computer identifies these assets under **Personnel Recovery**. These include:

CV-22 (Osprey), HH- 60G (Pave Hawk) - Combat Rescue

AC-130U - Gunship ground attack

EC-130H/P - Aerial Refuel

A-10 Close Air Support (Warthog)
- GBU-31-Joint direct attack munition(JDAM) Good against SAMs and Control Centers, Marginal against Ramps, Buildings, and Runways about 28% damage per attack
- AGM-65K- Good against SAMs and Radar, poor range
- Hydra 70- Air to ground missile, Use against Radar and SAMs
- AIM-9- Air to Air Missile, Short range, Heat seeking must be off center

VICTORY CONDITION SCORING (also see Appendix 4)**:**

Team's success will be measured primarily on successful search and recovery operations of friendly forces.

Appendix 3:
Force Packaging Tips

Force Packaging Defined: The integration of air and space systems, each performing different functions of air and space power, synergistically applied to achieve the desired effects of a mission. (In basic terms, using the right type(s) of aircraft for the specific type of mission while utilizing the principles of war.)

Principles of War

Objective: directing military operations toward a defined and attainable objective that contributes to strategic, operational, and tactical aims.

Mass: concentrating the effects of air and space power at the most advantageous place and time to achieve decisive results

Security: enhances our freedom of action by reducing the vulnerability of friendly forces.

Surprise: leveraging the security principle by attacking the enemy at a time, place, or in a manner for which they are not prepared.

Unity of Command: unity of command ensures concentration of effort for every objective under one responsible commander.

Offensive: the purpose of an offensive action is to seize, retain, and exploit the initiative.

Maneuver: places the enemy in a position of disadvantage through the flexible application of combat power in a multidimensional combat space.

Economy of Force: judicious employment and distribution of forces.

Simplicity: avoids unnecessary complexity in organizing, preparing, planning, and conducting military operations.

Developing the Mission Package

1. Determine what effect you want to achieve on the target you're given.

2. Choose the best weapon or combination of weapons to achieve that effect and examine compatible platforms that can effectively deliver that weapon on the target.

3. Clearly identify enemy threats and figure out strategies to minimize or neutralize those threats.

4. Effectively coordinate any necessary detached mission support for the mission package's routing (the right assets in the right place at the right time).

Force Packaging Common Terms

Battle Damage Assessment (BDA): the determination of effects achieved based on Intelligence Surveillance Reconnaissance (ISR). BDA can be obtained with satellite photos, UAVs, special operations forces, or by other aircraft in the strike package.

Rendezvous Point: the point where all the aircraft in the strike package get together prior to the strike.

Break Point: the point where the aircraft separate after they have completed the mission. A break point can also be called a Split or Scatter point.

Ingress Route: standard routing that your strike package will use to enter into enemy territory.

Egress Route: standard routing that you strike package will use to exit enemy territory.

Detached Support

Aerial Refueling (you can't fly without gas)

Navigation and Positioning (flying in the right areas at right altitudes, GPS guided munitions, etc.)

C3 (communications, command and control such as AWACS seeing the big picture of friendly and enemy air assets to vector in escorts to protect the package carrying out the mission)

Weather (so that we know what types of munitions to carry, what types of ISR assets to task for BDA, or what altitudes to fly at and so on)

ISR for mission planning, mission execution and mission BDA, and so on.

Combat Search and Rescue (CSAR) forces to rescue downed aircrews in case friendly aircraft are shot down

The following Steps Can Help You in Developing Packages.

1. Determine the objective and which targets you need to affect to achieve that objective.

2. Determine which systems can achieve those desired effects.

3. Determine the limitations of those systems.

4. Determine which systems can compensate for those limitations.

5. Look at the enemy threat and determine which systems can counter that threat (or, determine whether that threat is a target that needs to be affected by another package).

6. Think about support elements, while not a part of the strike packages, support packages are necessary for success.

Force Packaging Final Check

- Did you establish command and control among friendly forces and protect it if necessary?
- Knowing how important C2 is to us, what steps did you take to degrade the enemy's C2?
- How did you neutralize the enemy's air defenses? By physical attack, or are there other ways to neutralize a target without destroying it?
- How did you deny the enemy the use of air space, both theirs and ours?
- Did you consider denial on a temporary basis or did you seek permanent destruction?
- What methods did you employ to neutralize the enemy fighters in enemy territory?
- Did you consider any defense while you are on the offensive?
- Did you accurately employ and protect our tankers and other high value air assets?

Appendix 4:
AFEX Set-up / Execution Checklist

HARDWARE / SOFTWARE

Do you have a copy of the AFEX DVD? _____

Do you have seven (7) desktop or laptop computers (with mice)? _____

Have you loaded the AFEX program to each of the computers? _____

Have you created a shortcut to the map B-52 icon for each computer? _____

Have you selected a means of connecting the computers? _____

OPTION 1: Stand-along router with LAN cables _____

OPTION 2: MiFi or WiFi _____

OPTION 3: Local Area Network _____

Have you designated a host computer? _____

Have you obtained the IP address for the host computer? _____

PROCEDURES

Are you familiar with the steps to logging in as the host? _____

Are you familiar with the steps to logging in as a network caller? _____

Are you familiar with the steps to assigning aircraft to players? _____

EXECUTION STEPS

Are all players logged in? _____

Has the team provided the instructor an informal in-brief (optional)? _____

Has the host requested permission from the instructor to begin play in 2X mode? _____

Are players taking notes (lesson learned) throughout the process? _____

Has the host saved a copy of the victory condition report? _____

Have players prepared an informal debrief (optional)? _____